



Summary: Final-year Electronic Engineering student skilled in embedded systems, FPGA design, and AI-based hardware development. Driven by a strong ambition to advance intelligent electronic systems and contribute to next-generation AI-powered hardware solutions. Strong interest in signal processing and modern wireless communication systems, including adaptive beamforming techniques.

Matee ur Rasool Meladi

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GoogleScholar: <https://scholar.google.com/citations?user=ACApfRwAAAAJ&hl=en>

Technical Skills

- Python
- C++
- Verilog
- MATLAB
- Machine Learning
- Assembly Language
- RISC-V Architecture
- Sensor Deployment
- Sensor data acquisition
- IoT integration
- Model Deployment
- AI/Signal Processing

Software Experienced

- MATLAB
- Simulink
- VS Studio
- Xilinx ISE 14.7
- Cadence
- Virtuos
- Google Colab
- CST Studio
- Easy EDA

Education

Bachelor of Science (BS) in Electronics Engineering

The Islamia University of Bahawalpur (IUB), Pakistan

CGPA: **3.83 / 4.0** | Expected Graduation: **July 2027** ([Transcript](#))

Relevant Courses: Digital System Design, python, Oops, C++, Digital logic Design, Digital Signal Processing, Artificial Intelligence, Computer Architecture, Embedded System, Electromagnetic Theory, Linear Algebra, Electronics I & II, Power Electronics.

Research Experience

- **CARE LAB (CENTER OF AUTOMATION, ROBOTICS AND EMBEDDED SYSTEM) | AUGUST 2024-PRESENT**

Gained hands-on experience in IoT and embedded system design, focusing on real-time hardware integration and testing. Learned trouble-shooting techniques for microcontroller-based and sensor-driven system. Worked on hardware development projects, combining electronics, programming, and automation concepts for practical applications.

- **DEEP EMBEDDED LAB | NOV 2024 - AUGUST 2025**

The Islamia University of Bahawalpur

Completed an intensive AI workshops and training program, covering core concepts from fundamentals to advanced applications. Gained hands-on experience in machine learning, deep learning, and neural network implementation on embedded platforms. Developed practical projects integrating AI with hardware systems, enhance embedded solution.

- **COMMUNICATION LAB | MAR 2025 – JUN 2025**

The Islamia University of Bahawalpur

Worked on a CSI-based Human Activity Recognition (CSI-HAR) project, collecting raw Channel State Information data from multiple subjects performing different activities. Processed and analyzed the data using signal processing techniques and filtering methods to extract meaningful features. Trained machine learning models for activity classification and deployed the system on a Raspberry Pi for real-time activity recognition and edge-based implementation.

- MS Word
- Porteous
- Excel
- Blynk
- Streamlit
- Arduino IDE
- Desktop Application
- Embedded Systems Projects

Interest and Activities

- Keeping up-to-date with electronics and emerging technologies.
- Exploring AI, robotics, and intelligent automation systems.
- Interested in radar systems, unmanned aerial systems (UAVs), and defense technologies

Certification

- Completed the online course [Engineering and product design](#) authorized by Arizona State University.
- Also visits various industry tours. During this we explore industry equipment and working process by implement the electronics.
- Successfully completed the course AI for Everyone, authorized by Deep Learning. ([Certificate](#))

Languages

- **Urdu** — Native
- **Punjabi** — Native/Bilingual
- **English** — Working Proficiency

Projects:

FPGA Implementation of a 3-Bit Up/Down Counter Dec-2025

Tools Explored: Verilog HDL, Xilinx ISE 14.7, Spartan-3 FPGA, DSD. Designed and implemented a 3-bit synchronous up/down counter using Verilog HDL and deployed it on a Xilinx Spartan-3 FPGA. Verified the RTL design through functional simulation, synthesized the hardware, and implemented it on the FPGA with LED and seven-segment display outputs. Optimized the design using synchronous logic and hardware debugging to ensure reliable real-time operation.

PLC–ESP32 Integration for Industrial Automation Work cell ([Project](#))

Jun 2025 – Aug 2025

Tools Explored: ESP32, Industrial Automation, IoT Integration. Integrated a PLC Work cell 34-120 machine with an ESP32 controller at the Center of Automation, Robotics and Embedded Systems (CARE), enabling communication between traditional PLC systems and modern IoT controllers. The project explored industrial automation concepts and demonstrated how PLC systems can extend toward smart Industry 4.0 applications.

IoT-Based Smart Adaptive Speed and Safety Control for Electric Vehicle

Jan 2024 – Apr 2024

Tools Explored: C/C++, Microcontroller, Hardware Setup, Sensor Integration, IoT.

Designed and implemented an IoT-enabled system that automatically adjusts vehicle speed based on road and traffic conditions, enhancing operational safety and energy efficiency in electric vehicles.

Full-Stack School Management System ([Project](#))

Oct 2025 – Nov 2025

Tools Explored: React, Python, FastAPI, SQLAlchemy, REST API. Designed and developed a full-stack school management system with a React frontend and FastAPI backend to manage student records, fees, and payroll data securely. Implemented RESTful APIs, secure authentication using JWT and Bcrypt, and deployed the application using modern cloud platforms.

Achievements

- **Honhaar Scholarship Awardee** – Selected under the **Honhaar Scholarship Program** for outstanding academic performance and continued educational achievement.

References

Dr. Waqas Anjum Khichi

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Electronic Engineering
University College of Engineering &
Technology
The Islamia University of Bahawalpur,
Pakistan

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Expert in Linear Control System, Linear
Algebra, Electronics I & II, and IoT based
Engineering Application

Dr. Shahab Ahmad Niazi

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partment of Electronics Engineering The
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stan

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Expert in Computer Architecture, Com-
puter Network, Calculus, Digital Logic
Design

- **Prime Minister’s Laptop Scheme Awardee** – Received a laptop under the Prime Minister’s Laptop Scheme in recognition of academic performance and merit.

Publications:

- Matee ur Rasool, Abdul Salam, Ayesha Nazir, Talha Mushtaq.
“**Hybrid Attention-Guided Capsule Framework for Automated Facial
Acne Severity Classification.**”
in *Spectrum of Engineering Sciences*, vol. 3, no. 12, pp. 973–1002, 2025.
<https://www.thesesjournal.com/index.php/1/article/view/1751> **(Published)**